MIL-M-28787/248A(EC)
21 July 1981
SUPERSEDING
MIL-M-28787/248(EC)
22 January 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, NON-INVERTING, DIGITAL, KEY CODE AMC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic digital non-inverting buffer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.058 failures per million operating hours for class I or 0.131 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AMC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/248 -1 for class I and M28787/248 -2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage(V_{DD}) of 3.0 to 18.0 V ±.5 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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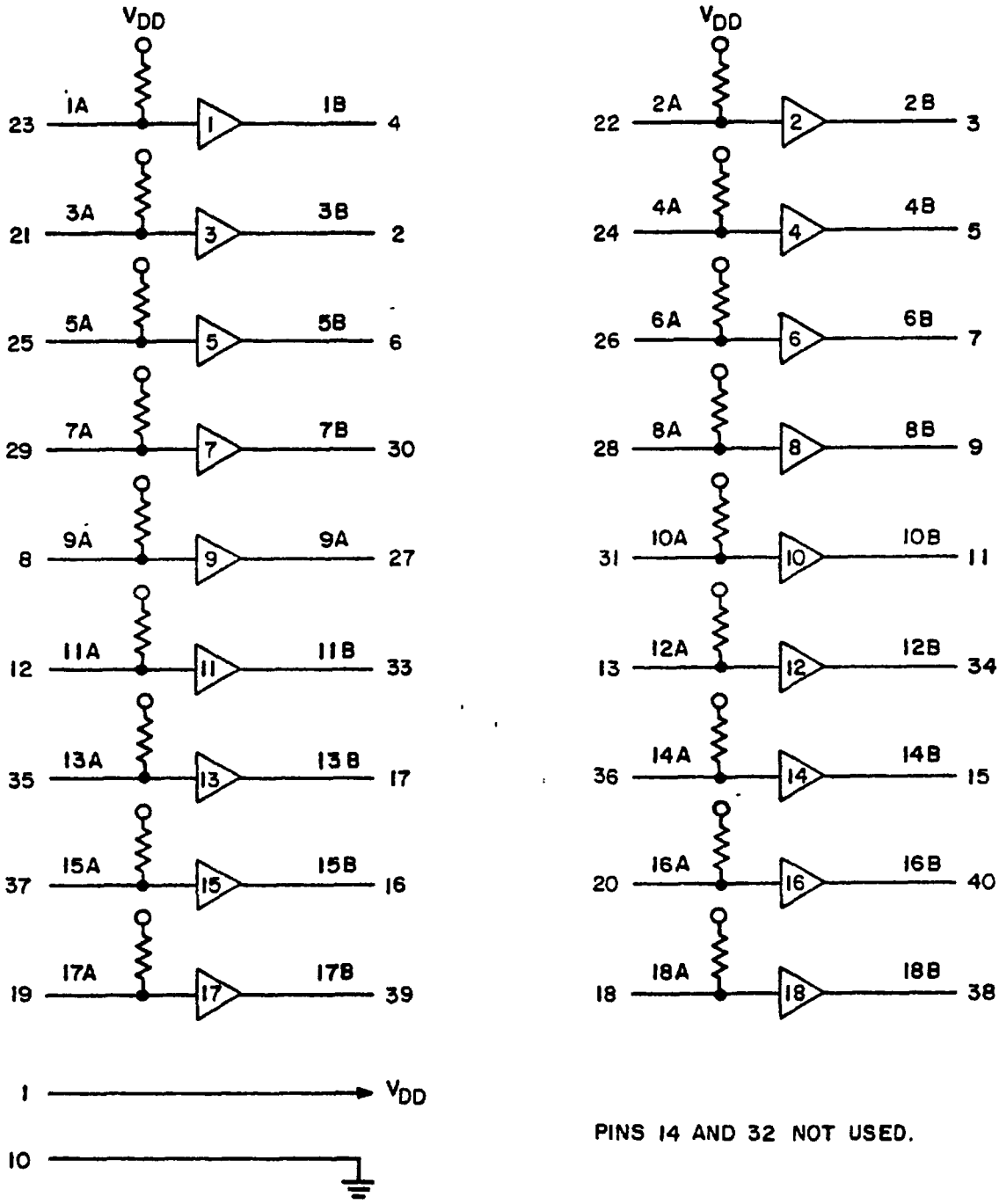


FIGURE 1. AMC module function diagram, noninverting buffer.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation Initial	II	S	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	II	S	1.0	-	1.0	-	1.0	-	1.0	-	1.0	-	MΩ
C1-C18	Negative input clamp diode	III	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Static output voltage, low level	IV	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Static output voltage, high level	V	X	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
F1-F18	Logic level conversion	VI	X	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
G1-G18	Static output voltage, low level (V_{DD} = 15V)	VII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
H1-H18	Static output voltage, high level (V_{DD} = 15V)	VIII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
J1-J18	Input current, high level	IX	X	0	200	0	200	0	200	---	---	0	200	nA
K1-K18	Input current, low level	X	X	-30	-45	-30	-45	-30	-45	---	---	-30	-45	µA
L1	Static power supply current, low level	XI	X	0	12	0	12	0	175	0	12	0	360	µA
L2	Static power supply current, high level	XI	X	0	12	0	12	0	175	0	12	0	360	µA
M1-M18	Transition time, t_{TLH}, (C_L = 50 pF)	XII	S	20	200	20	200	22	220	15	150	24	245	ns
M19-M36	Transition time, t_{TLH}, (C_L = 150 pF)	XII	Q	40	400	40	400	44	440	30	305	49	490	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N18	Transition time, t_{THL}, (C_L = 50 pF)	XIII	S	8	80	8	80	9	90	6	60	10	100	ns
N19-N36	Transition time, t_{THL}, (C_L = 150 pF)	XIII	Q	14	140	14	140	15	155	17	175	17	170	ns
P1-P18	Propagation delay time, t_{PLH}, (C_L = 50 pF)	XIV	S	16	160	16	160	17	175	12	120	19	195	ns
P19-P36	Propagation delay time t_{PLH}, (C_L = 150 pF)	XIV	Q	21	210	21	210	23	232	16	160	25	255	ns
Q1-Q18	Propagation delay time t_{PHL}, (C_L = 50 pF)	XV	S	11	110	11	110	12	120	8	85	13	135	ns
Q19-Q36	Propagation delay time t_{PHL}, (C_L = 150 pF)	XV	Q	13	135	13	135	15	150	10	105	16	165	ns
R1-R18	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XIX	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
V1-V18	Input capacitance	XX	Q	2	25	2	25	2	25	2	25	2	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through H, L through Q, and N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following life test and while still at the maximum class temperature, electrical tests C through H, L through Q, and U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum quiescent power dissipation	5.4	5.4	mW
Input current			
Low level (V_{IL}=0V, V_{DD}=18V	-45 max	-45 max	µA
High level (V_{IH}=18V, V_{DD}=18V)	200 max	200 max	nA

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Input voltage			
Low level			
V_{DD}=5.0 V -55°C	0 min, 1.5 max	0 min, 1.5 max	V
0° and 25°C	0 min, 1.5 max	0 min, 1.5 max	V
60°C	0 min, 1.5 max	0 min, 1.5 max	V
100°C	0 min, 1.5 max	0 min, 1.5 max	V
V_{DD} = 15.0V -55°C	0 min, 4.0 max	0 min, 4.0 max	V
0° and 25°C	0 min, 4.0 max	0 min, 4.0 max	V
60°C	0 min, 4.0 max	0 min, 4.0 max	V
100°C	0 min, 4.0 max	0 min, 4.0 max	V
High level			
V_{DD}=5.0 V -55°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
0° and 25°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
60°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
100°C	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
Output voltage (V_{DD}=5.0V)			
Low level (I_{OL}=3.75 mA at -55°C)	0.4 min	0.4 min	V
(I_{OL}=3.2 mA at 0° and 25°C)	0.4 min	0.4 min	V
(I_{OL}=2.68 mA at 60°C)	0.4 min	0.4 min	V
(I_{OL}=2.1 mA at 100°C)	0.4 min	0.4 min	V
High level (I_{OH}=-1.85 mA at -55°C)	2.5 min	2.5 min	V
(I_{OH}=-1.25 mA at 0° and 25°C)	2.5 min	2.5 min	V
(I_{OH}=-1.08 mA at 60°C)	2.5 min	2.5 min	V
(I_{OH}=-0.90 mA at 100°C)	2.5 min	2.5 min	V
Output voltage (V_{DD} = 15 V)			
Low level (I_{OL}=30 mA at -55°C)	1.5 min	1.5 min	V
(I_{OL}=24 mA at 0°C and 25°C)	1.5 min	1.5 min	V
(I_{OL}=20.6 mA at 60°C)	1.5 min	1.5 min	V
(I_{OL}=16.8 mA at 100°C)	1.5 min	1.5 min	V
High level (I_{OH}=-5.5 mA at -55°C)	13.5 min	13.5 min	V
(I_{OH}=-3.75 mA at 0° and 25°C)	13.5 min	13.5 min	V
(I_{OH}=-3.26 mA at 60°C)	13.5 min	13.5 min	V
(I_{OH}=-2.7 mA at 100°C)	13.5 min	13.5 min	V

* 6.5 Module function. This contains 18 independent non-inverting buffers. The module may be used as a logic-level shifter if V_{DD} is the same as that of the driven TTL/DTL circuit. The input-signal high (V_{IH}) can exceed the V_{DD} supply voltage when these devices are used for logic level conversion.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

6.5.2 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed.

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

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b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD}-V_{SS}) should not be applied as this will forward bias the input protection diodes.

e. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximately by the formula:

$$P_{Total} = (CV_{DD}^{2}f)N + \text{quiescent power dissipation.}$$

where C = load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

* 6.6 CAUTION: This caution applies to each test figure. Failure to comply with the following procedures will result in irreversable module damage.

a. All test equipment must be grounded.

b. V_{DD} shall be applied before applying input signals or conditioning voltages.

c. All inputs signals and conditioning voltages shall be removed before V_{DD} is disconnected.

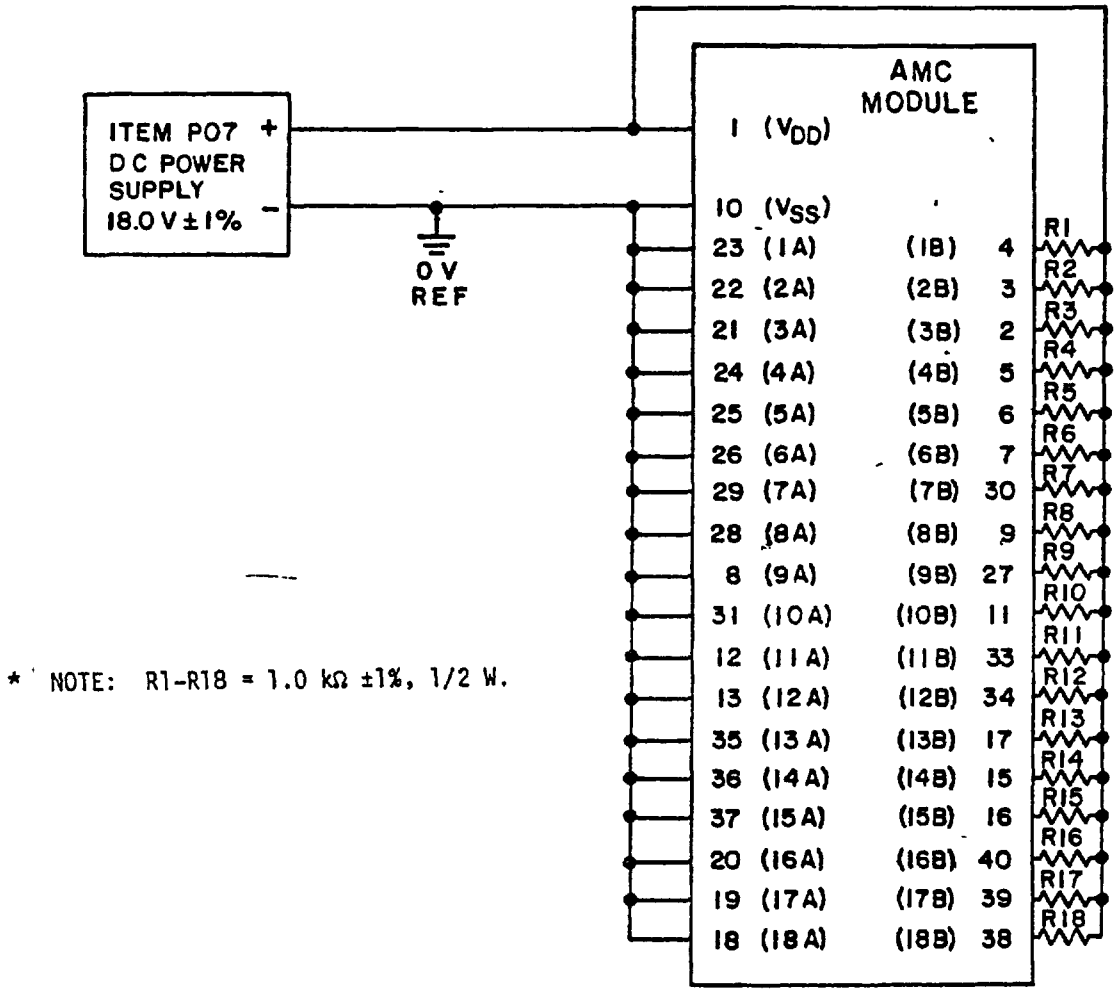
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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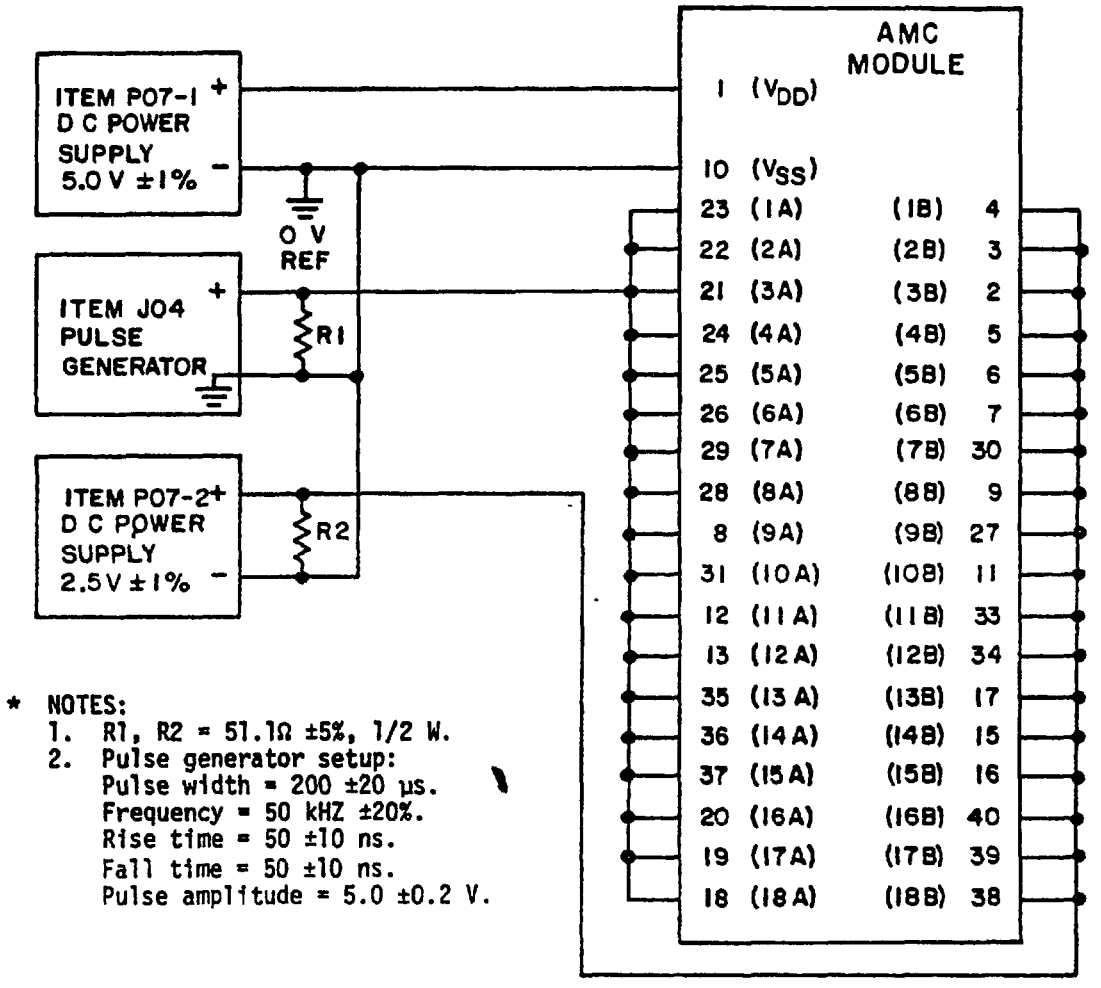
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* NOTE: R1-R18 = 1.0 kΩ ±1%, 1/2 W.

FIGURE 2. Operating temperature and life test circuit.

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* NOTES:
1. R1, R2 = 51.1Ω ±5%, 1/2 W.
2. Pulse generator setup:
   Pulse width = 200 ±20 µs.
   Frequency = 50 kHZ ±20%.
   Rise time = 50 ±10 ns.
   Fall time = 50 ±10 ns.
   Pulse amplitude = 5.0 ±0.2 V.

FIGURE 3. Component temperature test circuit.

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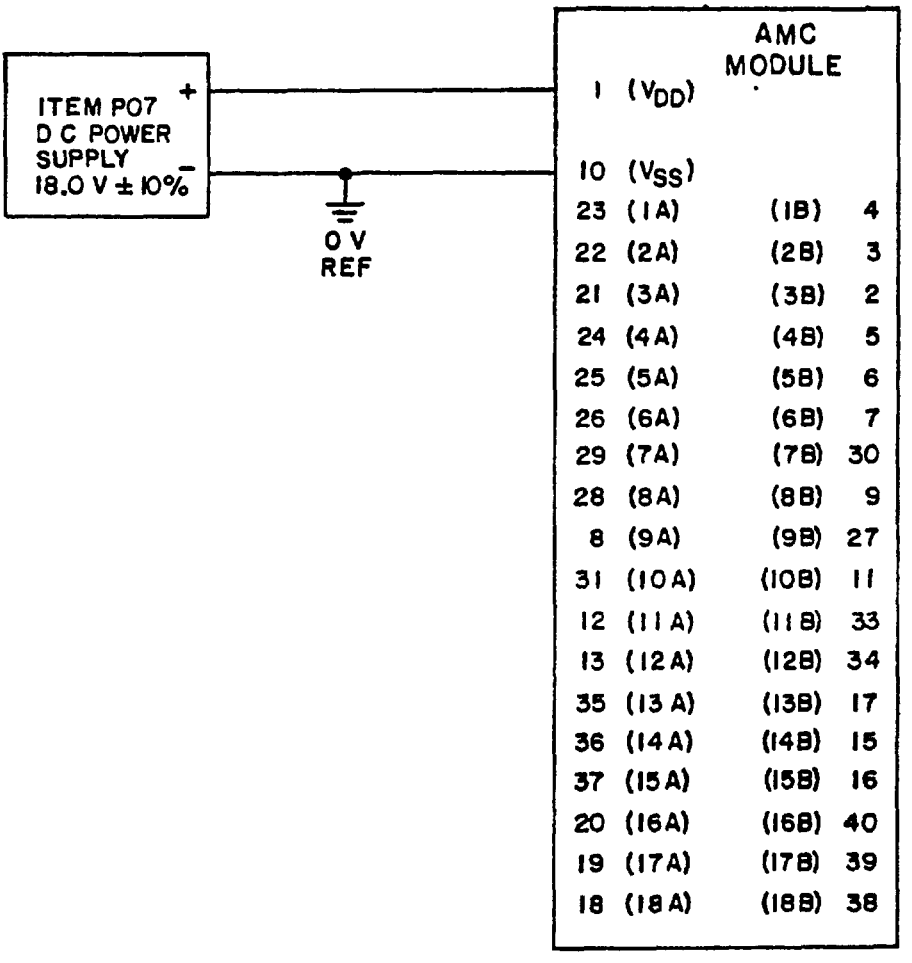


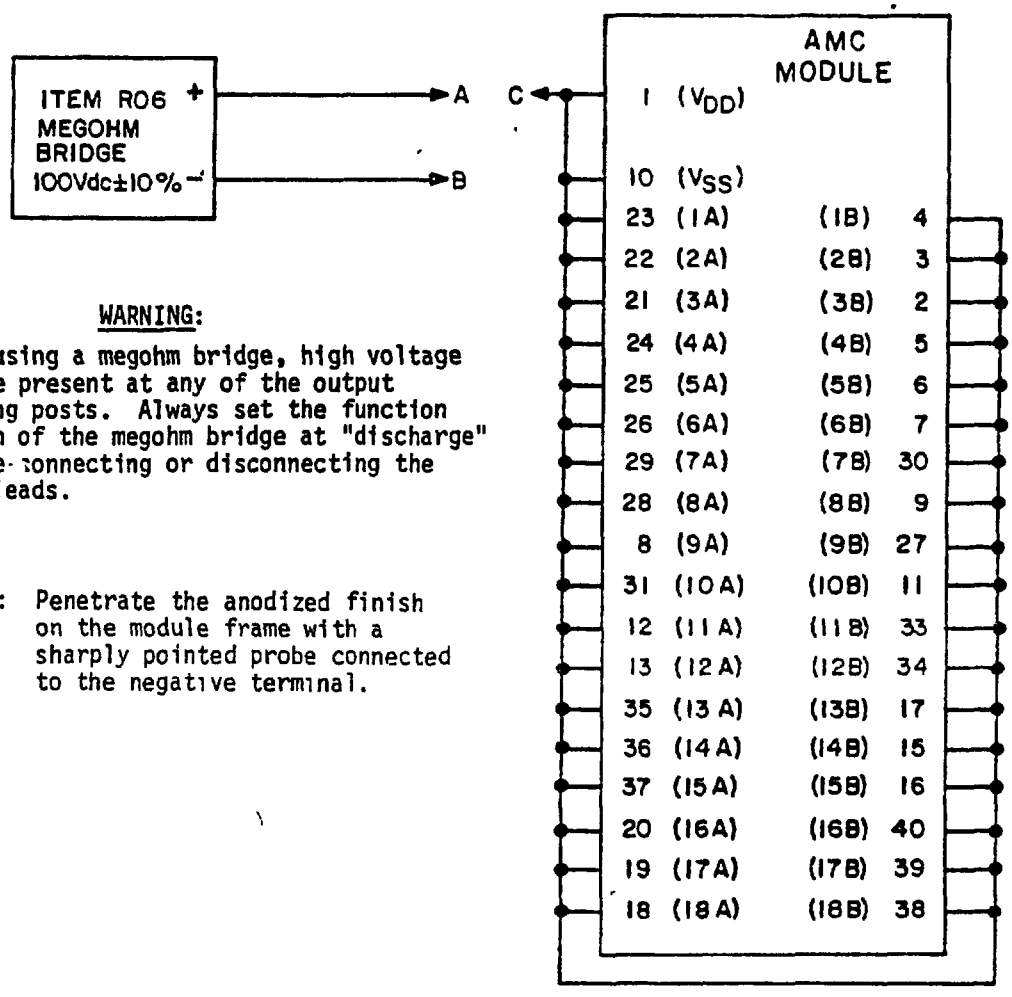
FIGURE 4. Vibration test circuit.

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* TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 14, 32	Frame	10	∞	MΩ	5
A2	Isolation	14, 32	C	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

* NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

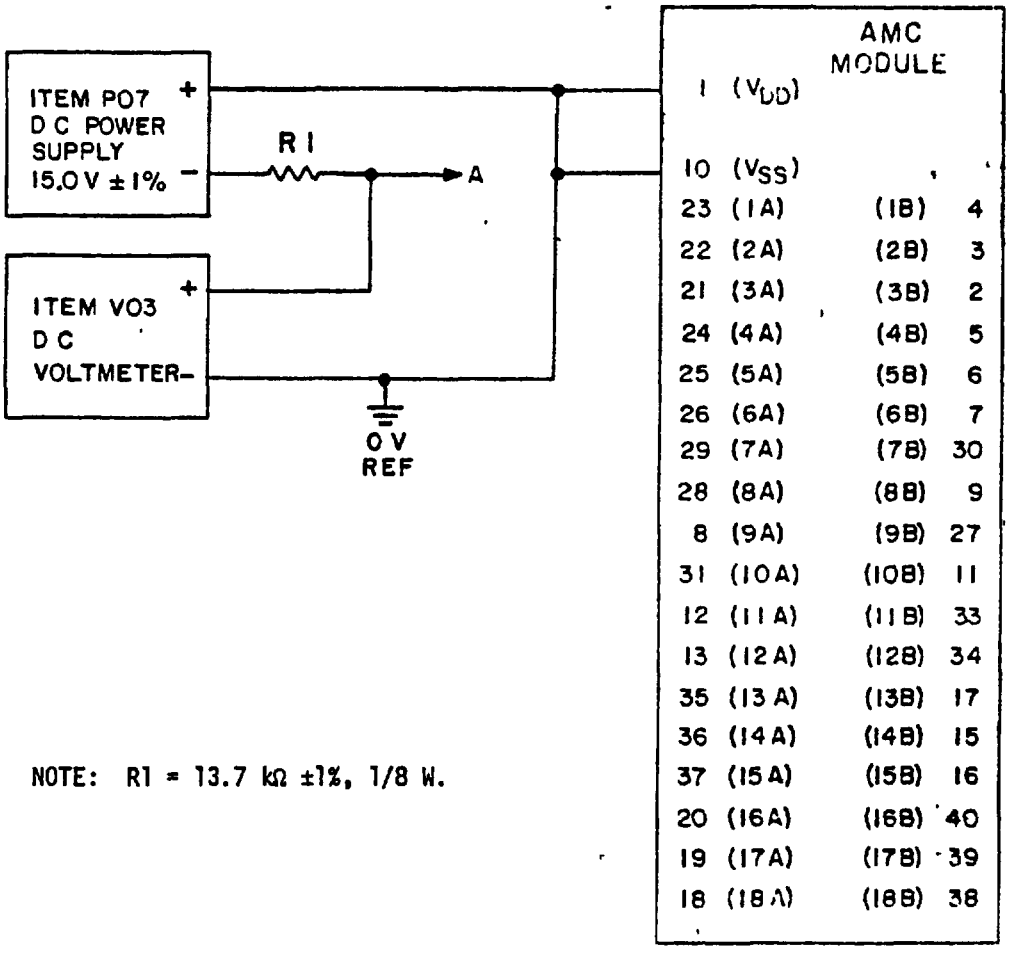
*FIGURE 5. Isolation test circuit for table II.

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* TABLE III. Negative input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1A	Negative input clamp diode voltage	23	-0.4	-1.5	V	6
C2	2A		22				
C3	3A		21				
C4	4A		24				
C5	5A		25				
C6	6A		26				
C7	7A		29				
C8	8A		28				
C9	9A		8				
C10	10A		31				
C11	11A		12				
C12	12A		13				
C13	13A		35				
C14	14A		36				
C15	15A		37				
C16	16A		20				
C17	17A		19				
C18	18A		18				

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NOTE: R1 = 13.7 kΩ ±1%, 1/8 W.

*FIGURE 6. Negative input clamp diode voltage test circuit for table III.

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TABLE IV. Static output voltage (low-level) test.

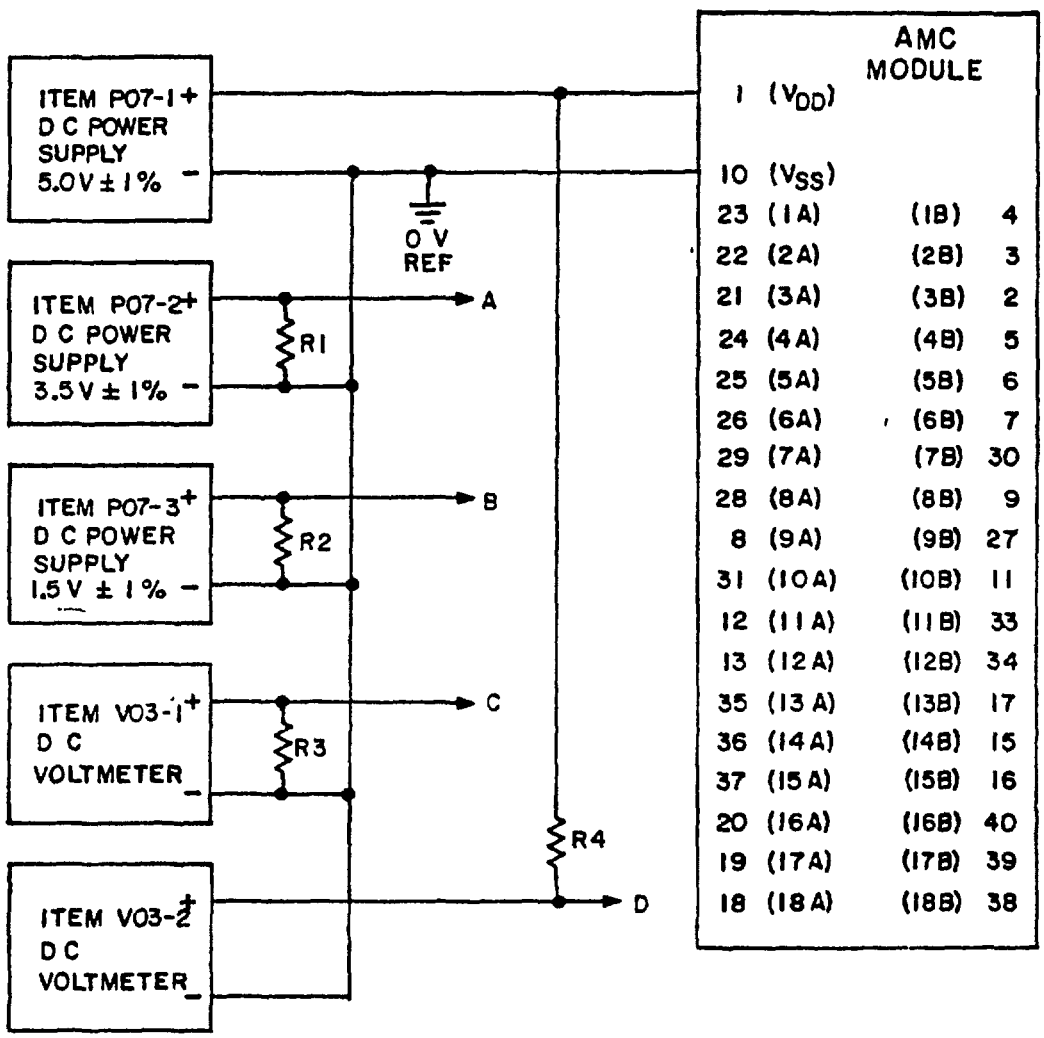
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	1B Static output voltage, low level	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	Open	4	0.0	0.4	V	7
D2	2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22		3				
D3	3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21		2				
D4	4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24		5				
D5	5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25		6				
D6	6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26		7				
D7	7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29		30				
D8	8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28		9				
D9	9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8		27				
D10	10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31		11				
D11	11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12		33				
D12	12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13		34				
D13	13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35		17				
D14	14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36		15				
D15	15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37		16				
D16	16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20		40				
D17	17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19		39				
D18	18B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18		38				

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TABLE V. Static output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1B Static output voltage, high level	23	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	4	Open	2.5	5.0	V	7
E2	2B	22	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3					
E3	3B	21	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2					
E4	4B	24	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5					
E5	5B	25	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6					
E6	6B	26	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7					
E7	7B	29	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30					
E8	8B	28	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9					
E9	9B	8	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	27					
E10	10B	31	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11					
E11	11B	12	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33					
E12	12B	13	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34					
E13	13B	35	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17					
E14	14B	36	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	15					
E15	15B	37	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	16					
E16	16B	20	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40					
E17	17B	19	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	39					
E18	18B	18	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,38,20,19	38					

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NOTES:
1. R1, R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = @ -55°C 1.37 kΩ ±1%, 1/8 W.
   @ 0° and 25°C 2.0 kΩ ±1%, 1/8 W.
   @ 60°C 2.32 kΩ ±1%, 1/8 W.
   @ 100°C 2.80 kΩ ±1%, 1/8 W.
3. R4 = @ -55°C 1.24 kΩ ±1%, 1/8 W.
   @ 0° and 25°C 1.43 kΩ ±1%, 1/8 W.
   @ 60°C 1.69 kΩ ±1%, 1/8 W.
   @ 100°C 2.21 kΩ ±1%, 1/8 W.

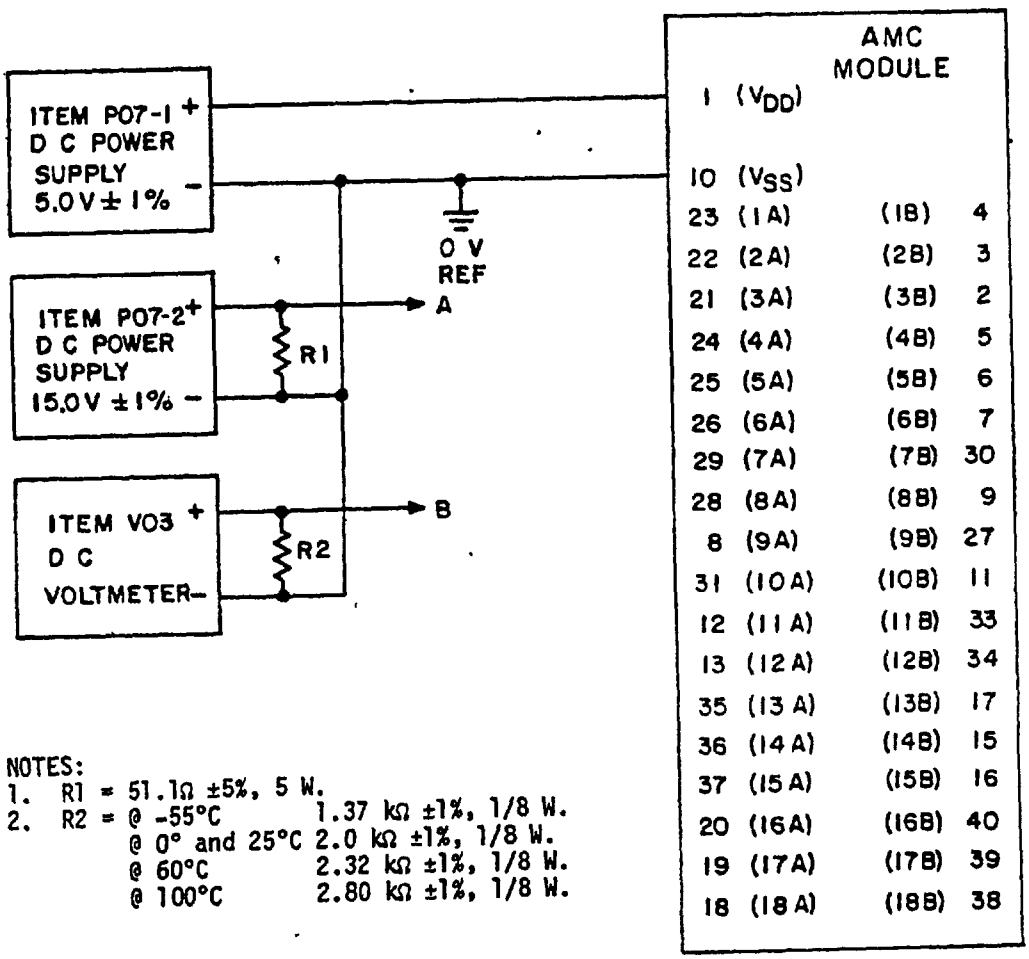
FIGURE 7. Static output voltage (low-level and high-level) test circuit for tables IV and V.

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TABLE VI. Logic level conversion test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	1B Logic level conversion	23,22,21,24,25,26,29, 28,8,31,12,13,35,36,37, 20,19,18	4	2.5	5.0	V	8
F2	2B		3				
F3	3B		2				
F4	4B		5				
F5	5B		6				
F6	6B		7				
F7	7B		30				
F8	8B		9				
F9	9B		27				
F10	10B		11				
F11	11B		33				
F12	12B		34				
F13	13B		17				
F14	14B		15				
F15	15B		16				
F16	16B		40				
F17	17B		39				
F18	18B		38				

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NOTES:
1. R1 = 51.1Ω ±5%, 5 W.
2. R2 = @ -55°C 1.37 kΩ ±1%, 1/8 W.
@ 0° and 25°C 2.0 kΩ ±1%, 1/8 W.
@ 60°C 2.32 kΩ ±1%, 1/8 W.
@ 100°C 2.80 kΩ ±1%, 1/8 W.

FIGURE 8. Logic level conversion test circuit for table VI.

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TABLE VII. Static output voltage (low-level) test.

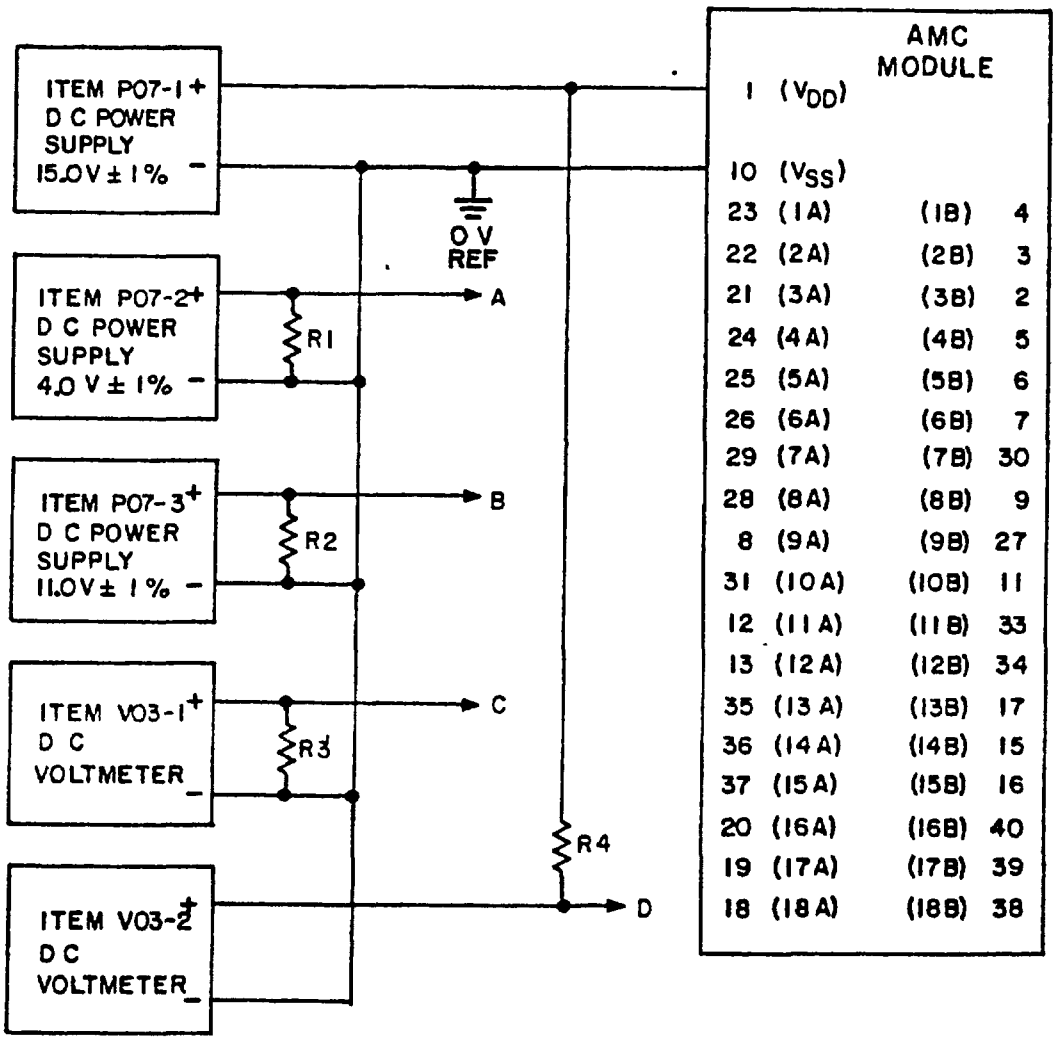
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1B	Static output voltage, low level(V_{DD}=15 V) 23,22,21,24,25,26,29,28,8,31,12,13,35,36,37,20,19,18	Open	Open	4	0.0	1.5	V	9
G2	2B				3				
G3	3B				2				
G4	4B				5				
G5	5B				6				
G6	6B				7				
G7	7B				30				
G8	8B				9				
G9	9B				27				
G10	10B				11				
G11	11B				33				
G12	12B				34				
G13	13B				17				
G14	14B				15				
G15	15B				16				
G16	16B				40				
G17	17B				39				
G18	18B				38				

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TABLE VIII. Static output voltage (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1B Static output voltage, high level (V_{DD}=15 V)	Open	23,22,21,24,25,26,29,28,8,31,12,13,35,36,37,20,19,18	4	Open	13.5	15.0	V	9
H2	2B			3					
H3	3B			2					
H4	4B			5					
H5	5B			6					
H6	6B			7					
H7	7B			30					
H8	8B			9					
H9	9B			27					
H10	10B			11					
H11	11B			33					
H12	12B			34					
H13	13B			17					
H14	14B			15					
H15	15B			16					
H16	16B			40					
H17	17B			39					
H18	18B			38					

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.
3. R3 = @ -55°C 2.43 kΩ ±1%, 1/8 W.
   @ 0° and 25°C 3.57 kΩ ±1%, 1/8 W.
   @ 60°C 4.12 kΩ ±1%, 1/8 W.
   @ 100°C 4.99 kΩ ±1%, 1/8 W.
4. R4 = @ -55°C 453Ω ±1%, 1/8 W.
   @ 0° and 25°C 562Ω ±1%, 1/8 W.
   @ 60°C 649Ω ±1%, 1/8 W.
   @ 100°C 806Ω ±1%, 1/8 W.

FIGURE 9. Static output voltage (low-level and high-level) test circuit for tables VII and VIII.

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TABLE IX. Input current (high-level) test.

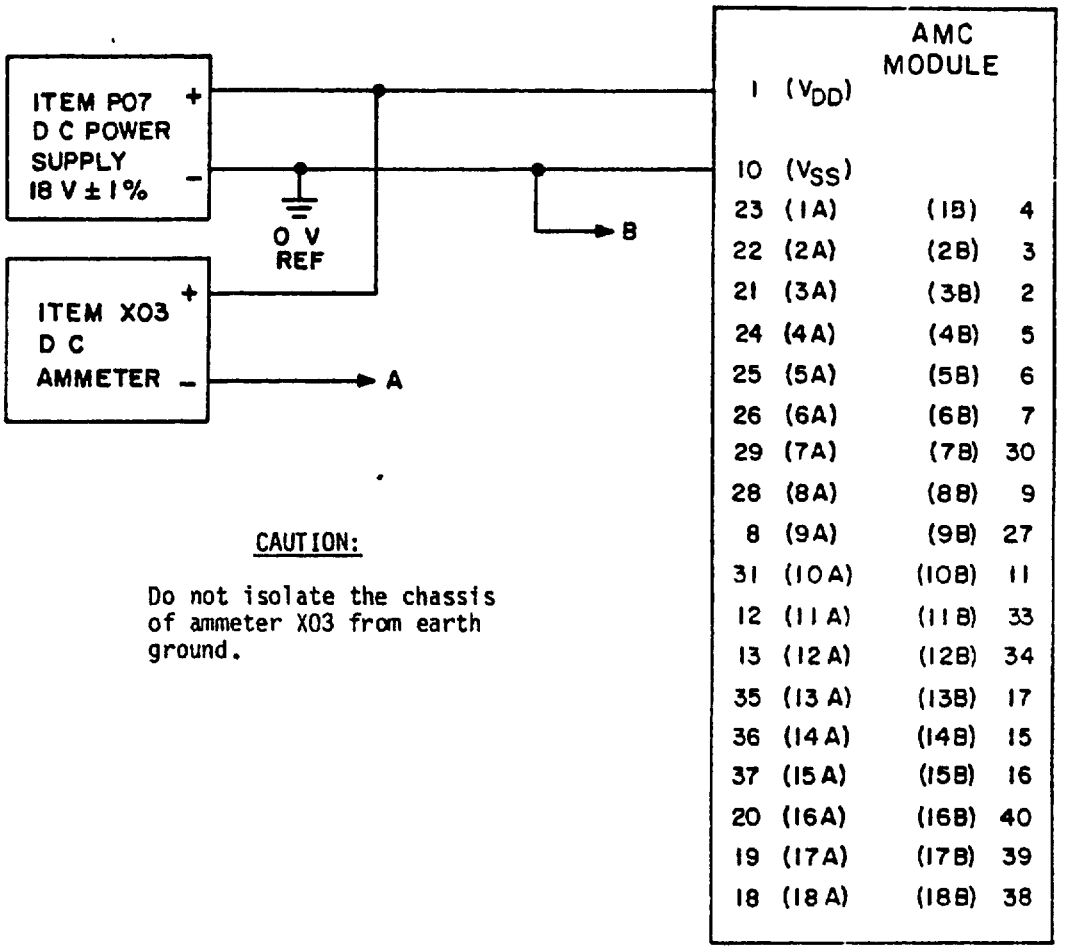
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
J1	1A Input current, high level	23	22,21,24,25,26,29,28,8,31,12,13,35,36,37,20,19,18	0	200	nA	10
J2	2A	22	23,21,24,25,26,29,28,8,31,12,13,35,36,37,20,19,18				
J3	3A	21	23,22,24,25,26,29,28,8,31,12,13,35,36,37,20,19,18				
J4	4A	24	23,22,21,25,26,29,28,8,31,12,13,35,36,37,20,19,18				
J5	5A	25	23,22,21,24,26,29,28,8,31,12,13,35,36,37,20,19,18				
J6	6A	26	23,22,21,24,25,29,28,8,31,12,13,35,36,37,20,19,18				
J7	7A	29	23,22,21,24,25,26,28,8,31,12,13,35,36,37,20,19,18				
J8	8A	28	23,22,21,24,25,26,29,8,31,12,13,35,36,37,20,19,18				
J9	9A	8	23,22,21,24,25,26,29,28,31,12,13,35,36,37,20,19,18				
J10	10A	31	23,22,21,24,25,26,29,28,8,12,13,35,36,37,20,19,18				
J11	11A	12	23,22,21,24,25,26,29,28,8,,31,13,35,36,37,20,19,18				
J12	12A	13	23,22,21,24,25,26,29,28,8,31,12,35,36,37,20,19,18				

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* TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J13	13A Input current, high level	35	23,22,21,24,25,26,29, 28,8,31,12,13,36,37,20, 19,18	0	200	nA	10
J14	14A	36	23,22,21,24,25,26,29, 28,8,31,12,13,35,37,20, 19,18				
J15	15A	37	23,22,21,24,25,26,29, 28,8,31,12,13,35,36,20, 19,18				
J16	16A	20	23,22,21,24,25,26,29, 28,8,31,12,13,35,36, 19,18				
J17	17A	19	23,22,21,24,25,26,29, 28,8,31,12,13,35,36,37, 20,19				
J18	18A	18	23,22,21,24,25,26,29, 28,8,31,12,13,35,36,37, 20,19				

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CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

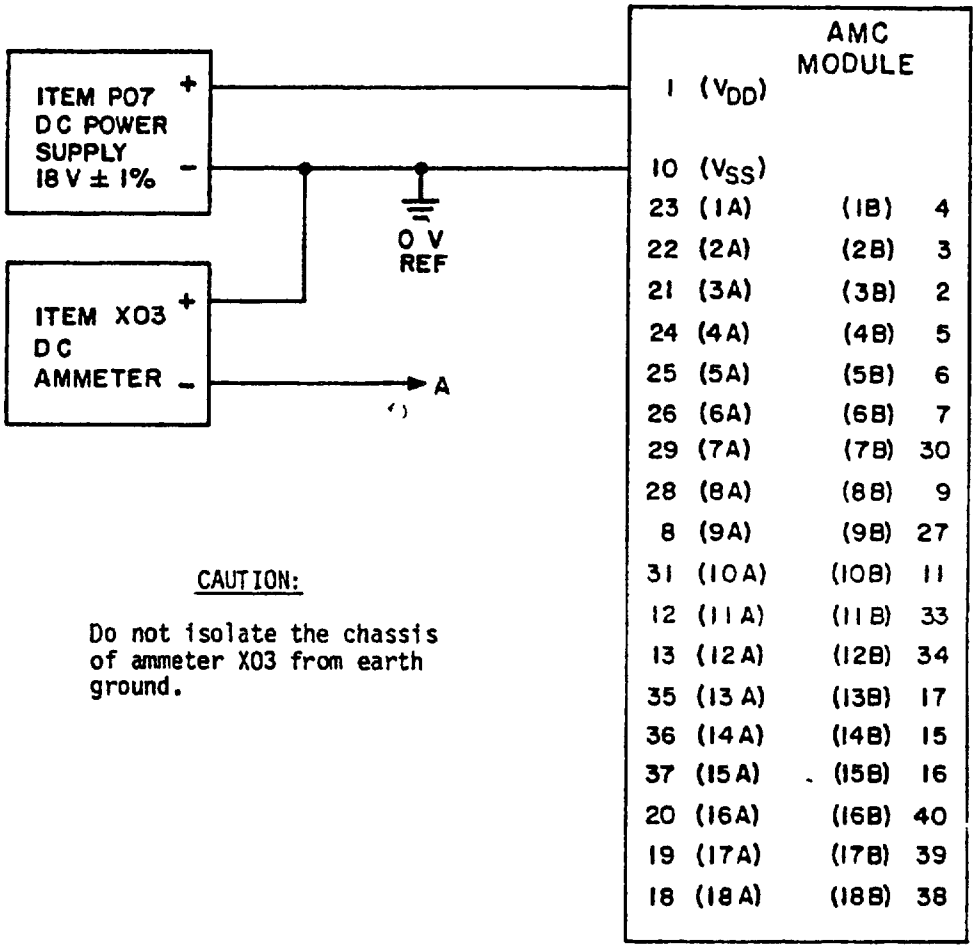
FIGURE 10. Input current (high-level) test circuit for table IX.

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TABLE X. Input current (low-level) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	1A	Input current, low level	23	-30	-45	μA	11
K2	2A	↓	22	↓	↓	↓	↓
K3	3A		21				
K4	4A		24				
K5	5A		25				
K6	6A		26				
K7	7A		29				
K8	8A		28				
K9	9A		8				
K10	10A		31				
K11	11A		12				
K12	12A		13				
K13	13A		35				
K14	14A		36				
K15	15A		37				
K16	16A		20				
K17	17A		19				
K18	18A		18				

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CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

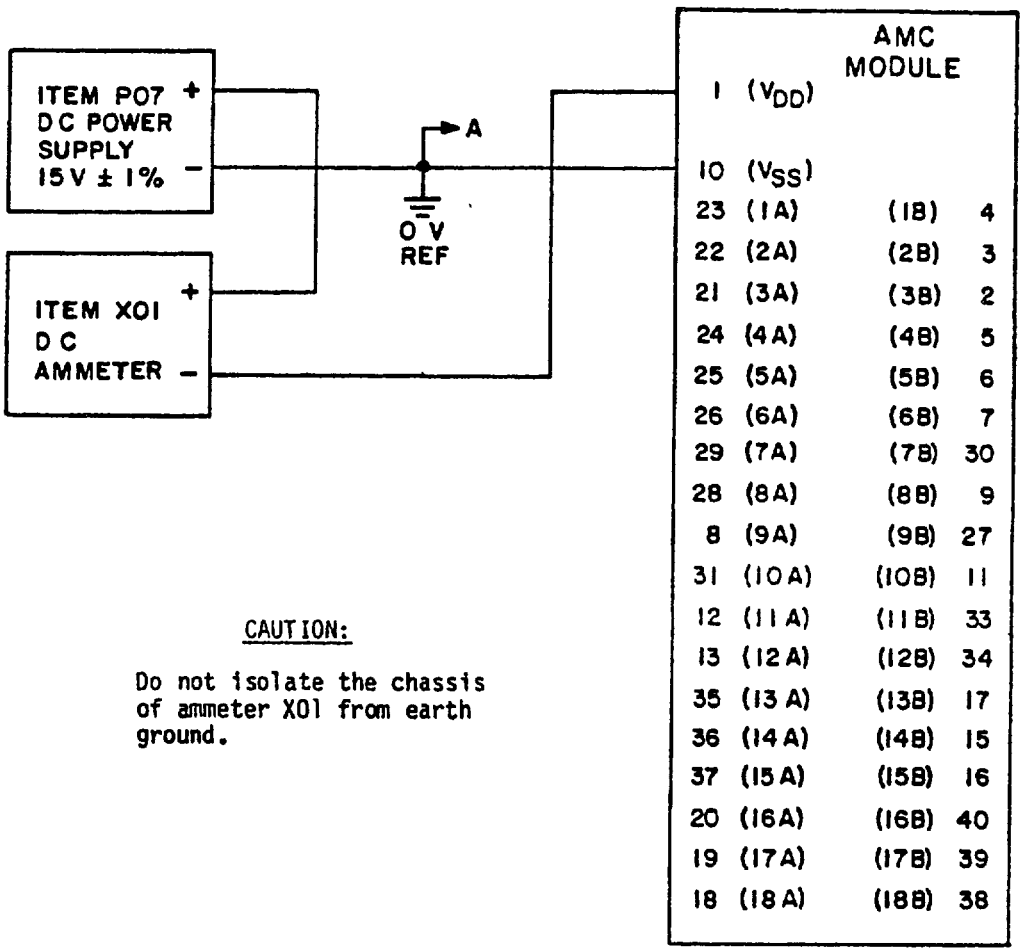
FIGURE 11. Input current (low-level) test circuit for table X.

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TABLE XI. Static power supply current (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	Static power supply current, low level	23,22,21,24,25,26,29,28, 8,31,12,13,35,36,37,20, 19,18	0.0	12.0	µA	12
L2	Static power supply current, high level	Open	0.0	12.0	µA	12

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 12. Static power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	1A to 1B Transition time, t_{TLH} (C_L=50 pF)	22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	23	4	20	200	ns	13, 14
M2	2A to 2B	23,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	22	3				
M3	3A to 3B	23,22,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	21	2				
M4	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
M5	5A to 5B	23,22,21,24,26,29,28,8, 31,12,13,35,36,37,20,19,18	25	6				
M6	6A to 6B	23,22,21,24,25,29,28,8, 31,12,13,35,36,37,20,19,18	26	7				
M7	7A to 7B	23,22,21,24,25,26,28,8, 31,12,13,35,36,37,20,19,18	29	30				
M8	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
M9	9A to 9B	23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
M10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
M11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
M12	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
M13	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
M14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
M15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
M16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
M17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				

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TABLE XII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M18	18A to 18B Transition time, t_{TLH} (C_L=50 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	20	200	ns	13, 14
M19	1A to 1B Transition time, t_{TLH} (C_L=150 pF)	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	40	400		
M20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
M21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
M22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
M23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
M24	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
M25	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
M26	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
M27	9A to 9B	23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
M28	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
M29	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
M30	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
M31	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
M32	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
M33	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				

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TABLE XII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M34	16A to 16B Transition time, t_{TLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40	40	400	ns	13, 14
M35	17A to 17B Transition time, t_{TLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39	40	400	ns	13, 14
M36	18A to 18B Transition time, t_{TLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	40	400	ns	13, 14

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1A to 1B Transition time, t_{THL} (C_L=50 pF)	22,21,24,25,26,29,28,8,31, 13,12,35,36,37,20,19,18	23	4	8	80	ns	13, 14
N2	2A to 2B	23,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	22	3				
N3	3A to 3B	23,22,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	21	2				
N4	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
N5	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
N6	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
N7	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
N8	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
N9	9A to 9B	23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
N10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
N11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
N12	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
N13	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
N14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
N15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
N16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
N17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				

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TABLE XIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M18	18A to 18B Transition time, t_{THL} (C_L=50 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	8	80	ns	13, 14
M19	1A to 1B Transition time, t_{THL} (C_L=150 pF)	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	14	140		
M20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
M21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
M22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
M23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
M24	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
M25	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
M26	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
M27	9A to 9B	23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
M28	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
M29	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
M30	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
M31	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
M32	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
M33	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				

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TABLE XIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N34	16A to 16B Transition time, t_{THL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40	14	140	ns	13, 14
N35	17A to 17B Transition time, t_{THL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39	14	140	ns	13, 14
N36	18A to 18B Transition time, t_{THL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	14	140	ns	13, 14

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TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	1A to 1B	Propagation delay time, t_{PLH} (C_L=50 pF)	22,21,24,25,26,29,28,8,31, 13,12,35,36,37,20,19,18	23	4	16	160	ns	13, 14
P2	2A to 2B		23,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	22	3				
P3	3A to 3B		23,22,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	21	2				
P4	4A to 4B		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
P5	5A to 5B		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
P6	6A to 6B		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
P7	7A to 7B		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
P8	8A to 8B		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
P9	9A to 9B		23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
P10	10A to 10B		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
P11	11A to 11B		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
P12	12A to 12B		23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
P13	13A to 13B		23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
P14	14A to 14B		23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
P15	15A to 15B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
P16	16A to 16B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
P17	17A to 17B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P18	18A to 18B	Propagation delay time, t_{PLH} (C_L=50 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	16	160	ns	13, 14
P19	1A to 1B	Propagation delay time, t_{PLH} (C_L=150 pF)	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	21	210		
P20	2A to 2B		23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
P21	3A to 3B		23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
P22	4A to 4B		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
P23	5A to 5B		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
P24	6A to 6B		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
P25	7A to 7B		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
P26	8A to 8B		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
P27	9A to 9B		23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
P28	10A to 10B		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
P29	11A to 11B		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
P30	12A to 12B		23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
P31	13A to 13B		23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
P32	14A to 14B		23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
P33	15A to 15B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				

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TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P34	16A to 16B	Propagation delay time, t_{PLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40	21	210	ns	13, 14
P35	17A to 17B	Propagation delay time, t_{PLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39	21	210	ns	13, 14
P36	18A to 18B	Propagation delay time, t_{PLH} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	21	210	ns	13, 14

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1B	Propagation delay time, t_{PHL} (C_L=50 pF)	22,21,24,25,26,29,28,8,31, 13,12,35,36,37,20,19,18	23	4	11	110	ns	13,14
Q2	2A to 2B		23,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	22	3				
Q3	3A to 3B		23,22,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	21	2				
Q4	4A to 4B		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
Q5	5A to 5B		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
Q6	6A to 6B		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
Q7	7A to 7B		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
Q8	8A to 8B		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
Q9	9A to 9B		23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
Q10	10A to 10B		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
Q11	11A to 11B		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
Q12	12A to 12B		23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
Q13	13A to 13B		23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
Q14	14A to 14B		23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
Q15	15A to 15B		23,22,21,24,25,26,29,28,31, 12,13,35,36,20,19,18	37	16				
Q16	16A to 16B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
Q17	17A to 17B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

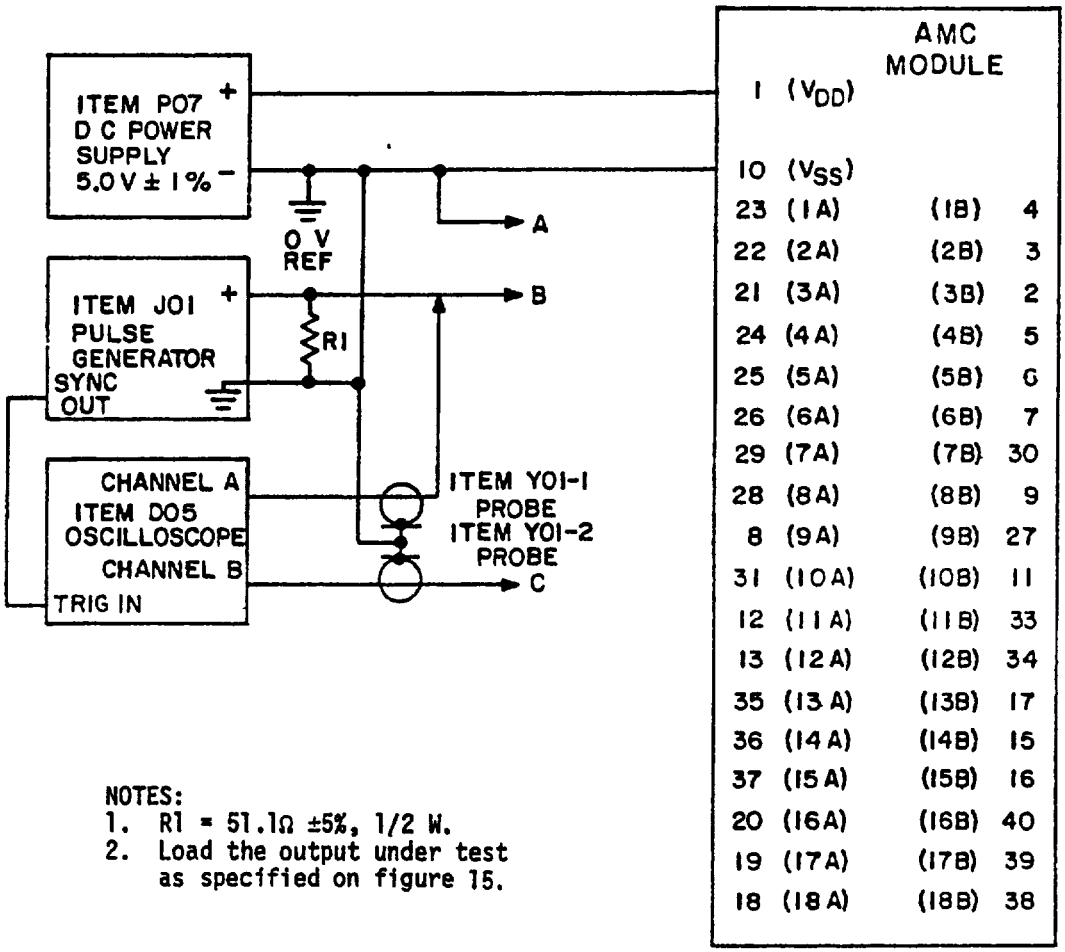
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q18	18A to 18B	Propagation delay time, t_{PHL} (C_L=50 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	11	110	ns	13, 14
Q19	1A to 1B	Propagation delay time, t_{PHL} (C_L=150 pF)	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	13	135		
Q20	2A to 2B		23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
Q21	3A to 3B		23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
Q22	4A to 4B		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
Q23	5A to 5B		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
Q24	6A to 6B		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
Q25	7A to 7B		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
Q26	8A to 8B		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
Q27	9A to 9B		23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	8	27				
Q28	10A to 10B		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
Q29	11A to 11B		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
Q30	12A to 12B		23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
Q31	13A to 13B		23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
Q32	14A to 14B		23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
Q33	15A to 15B		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q34	16A to 16B	Propagation delay time, t_{PHL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40	13	135	ns	13, 14
Q35	17A to 17B	Propagation delay time, t_{PHL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39	13	135	ns	13, 14
Q36	18A to 18B	Propagation delay time, t_{PHL} (C_L=150 pF)	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	13	135	ns	13, 14

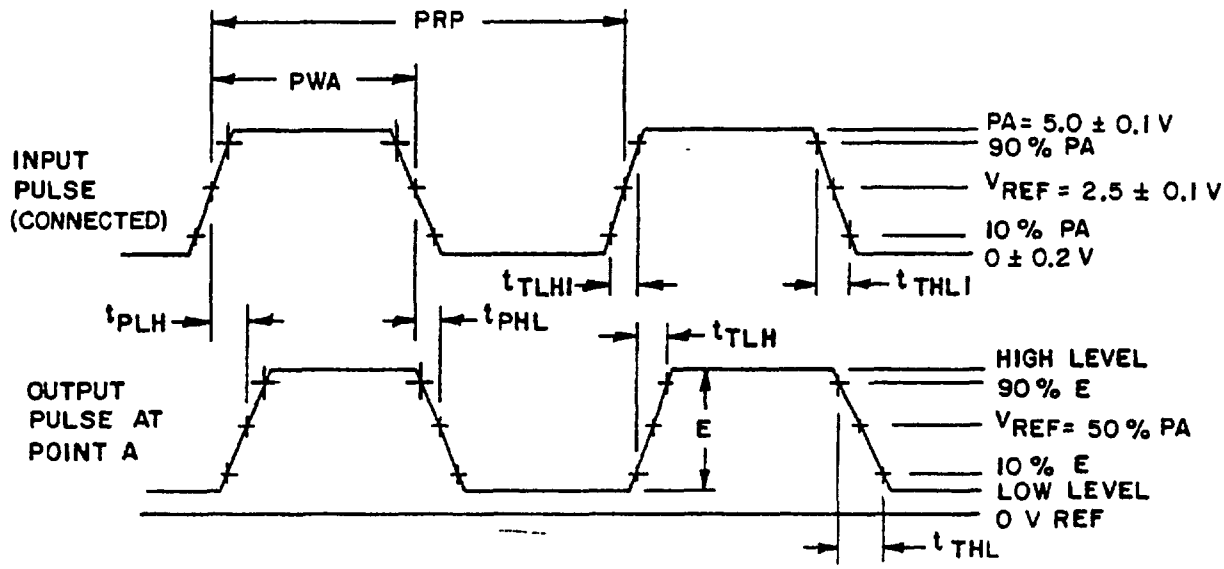
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load the output under test as specified on figure 15.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV.

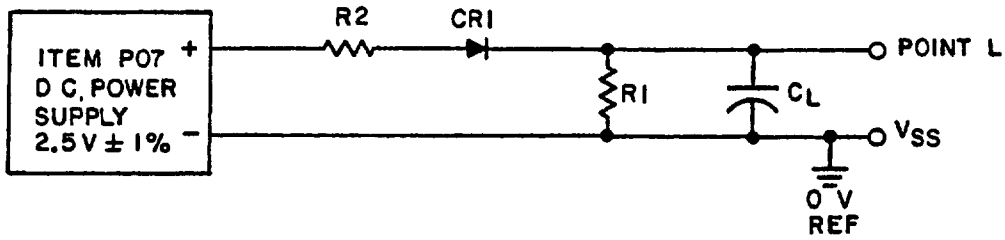
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 50.0 ±5 µs.
   PRP = 100.0 ±10 µs.
2. Before making the measurements specified in table XV, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 14. Transition time and propagation delay time test circuit waveforms for tables XII, XIII, XIV, and XV.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R2 = Select a value for current flow of 2.1 mA out of point L with point L held at 0.4 Vdc.
3. R1 = 2.80 kΩ ±1%, 1/8 W.
4. For test conditions M1-M8, N1-N18, P1-P18, Q1-Q18, R1-R18, and S1-S18, C_L = 50 ±5 pF (including probe and parasitic capacitance).
   For test conditions M19-M36, N19-N36, P19-P36, and Q19-Q36, C_L = 150 ±15 pF (including probe and parasitic capacitance).
5. The maximum lead length from point L to the appropriate module pin shall be less than 1.00 inch (25.4 mm).

FIGURE 15. Logic load circuit.

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TABLE XVI. Output crosstalk (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits -25°±5°C ambient: Min	Initial limits -25°±5°C ambient: Max	Unit	Test figure
R1	1B Output crosstalk, low level	23	Open	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	4	0	400	mV	16, 17
R2	2B	22		23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3				
R3	3B	21		23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2				
R4	4B	24		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5				
R5	5B	25		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6				
R6	6B	26		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7				
R7	7B	29		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30				
R8	8B	28		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9				
R9	9B	8		23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	27				
R10	10B	31		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11				
R11	11B	12		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33				
R12	12B	13		23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34				
R13	13B	35		23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17				

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TABLE XVI. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R14	14B	Output crosstalk, low level	36	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	15	0	400	mV	16, 17
R15	15B	↓	37	↓	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	16	↓	↓	↓	↓
R16	16B	↓	20	↓	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40	↓	↓	↓	↓
R17	17B	↓	19	↓	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	39	↓	↓	↓	↓
R18	18B	↓	18	↓	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	38	↓	↓	↓	↓

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TABLE XVII. Output crosstalk (high-level) test.

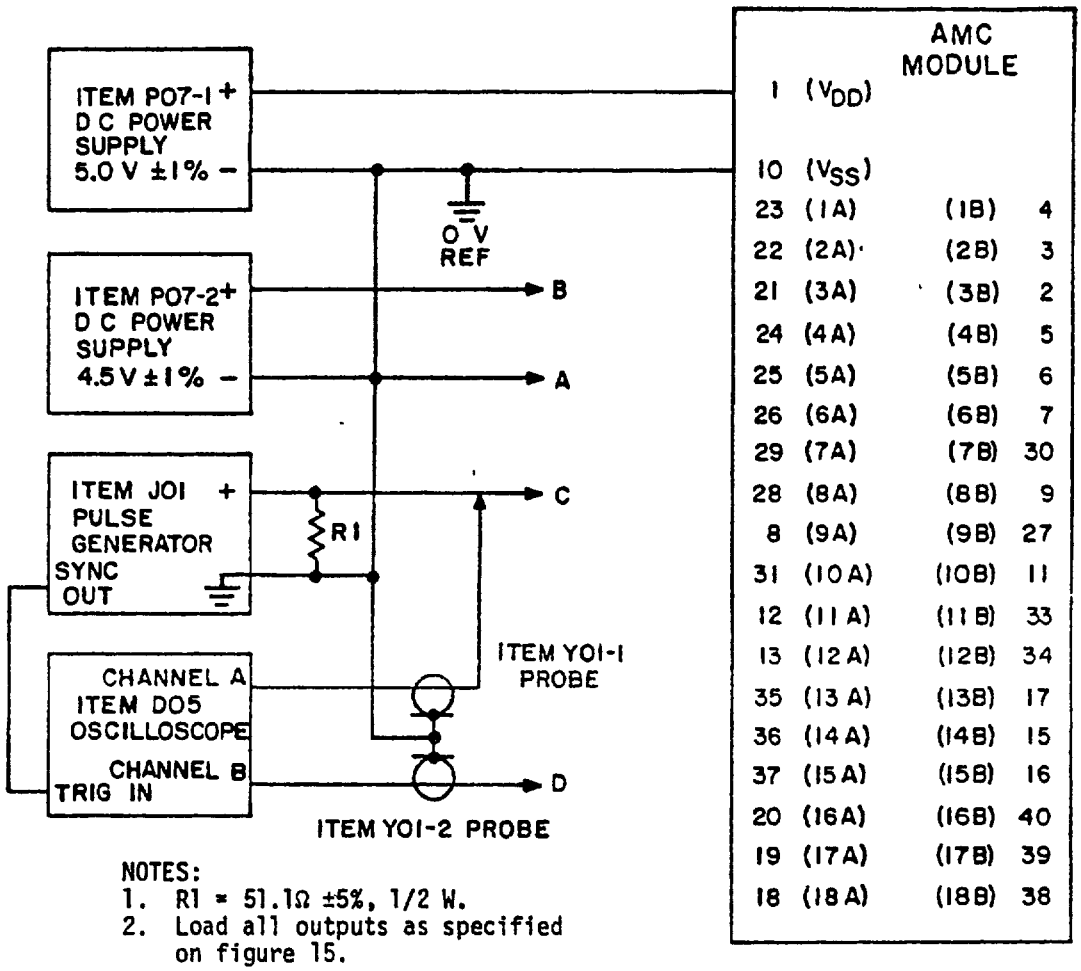
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
S1	1B Output crosstalk, high level	Open	23	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	4	0	400	mV	16,17
S2	2B		22	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3				
S3	3B		21	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2				
S4	4B		24	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5				
S5	5B		25	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6				
S6	6B		26	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7				
S7	7B		29	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30				
S8	8B		28	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9				
S9	9B		8	23,22,21,24,25,26,29,28,31, 12,13,35,36,37,20,19,18	27				
S10	10B		31	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11				
S11	11B		12	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33				
S12	12B		13	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34				
S13	13B		35	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17				

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TABLE XVII. Output crosstalk (high-level) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S14	14B	Output crosstalk, high level	Open	36	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	15	0	400	mV	16,17
S15	15B	↓	↓	37	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	16	↓	↓	↓	↓
S16	16B			20	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40				
S17	17B			19	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	39				
S18	18B			18	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	38				

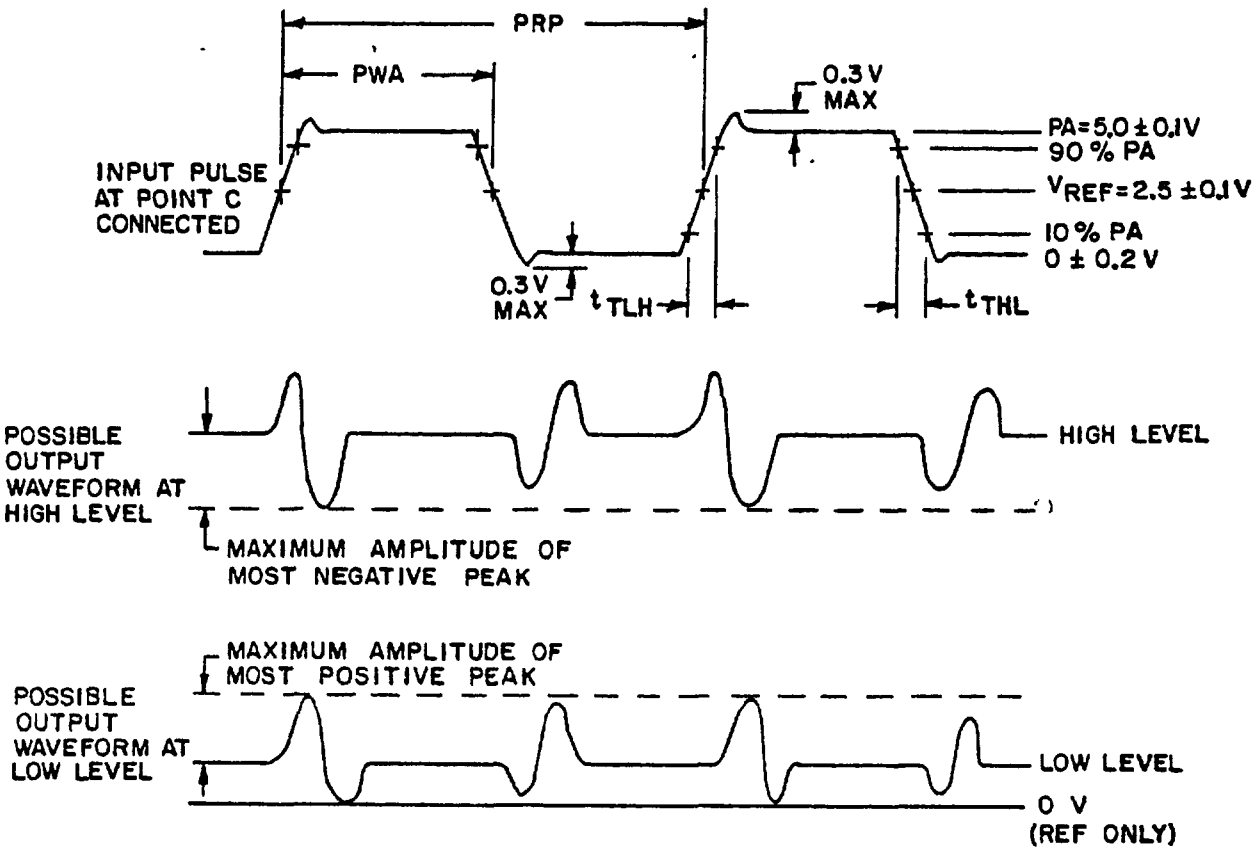
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load all outputs as specified on figure 15.

FIGURE 16. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

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NOTES:
1. Input pulse parameters. For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude. Pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.

   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
   PWA = 25 ±2.5 µs.
   PRP = 50.0 ±5.0 µs.

2. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1, Y01-2 to input and output pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 1.00 inch (25.4 mm).
3. Verify that the output low and high level are $\leq$ 0.4 V and $\geq$ 2.5 V respectively.

FIGURE 17. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	18, 19

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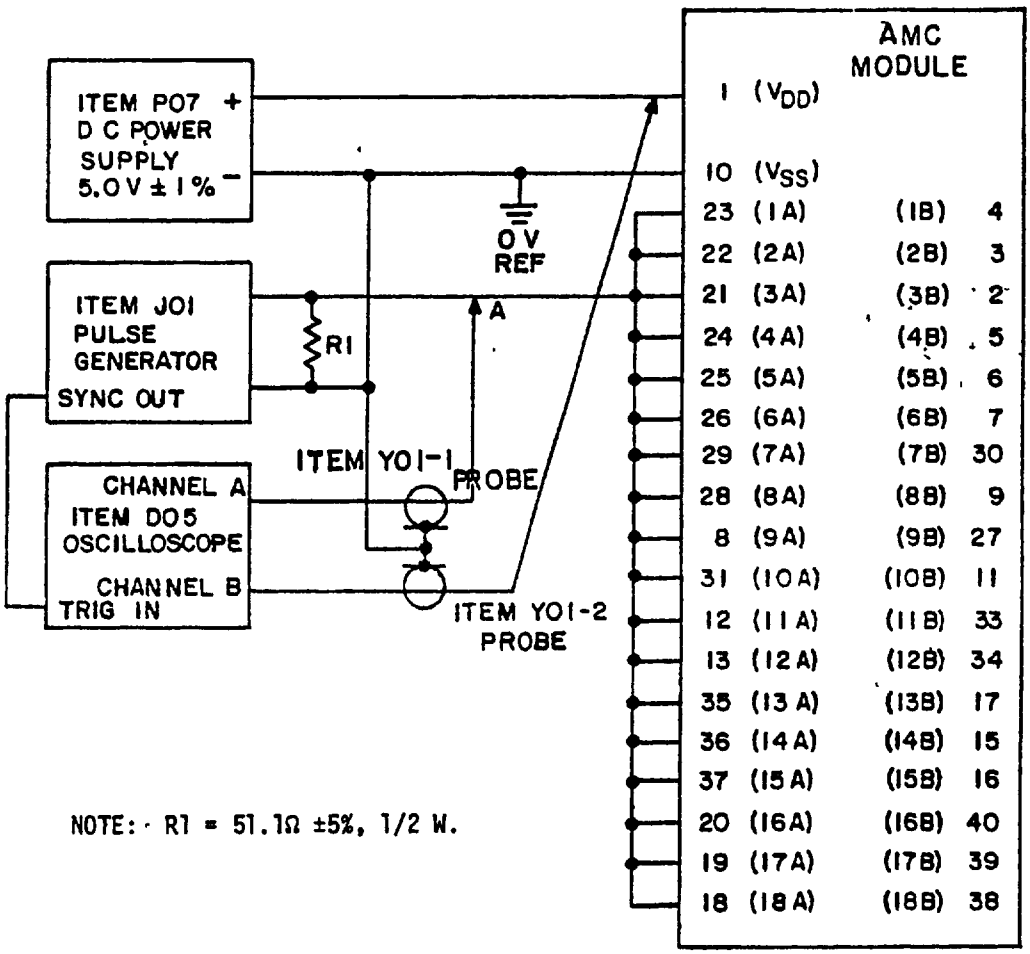
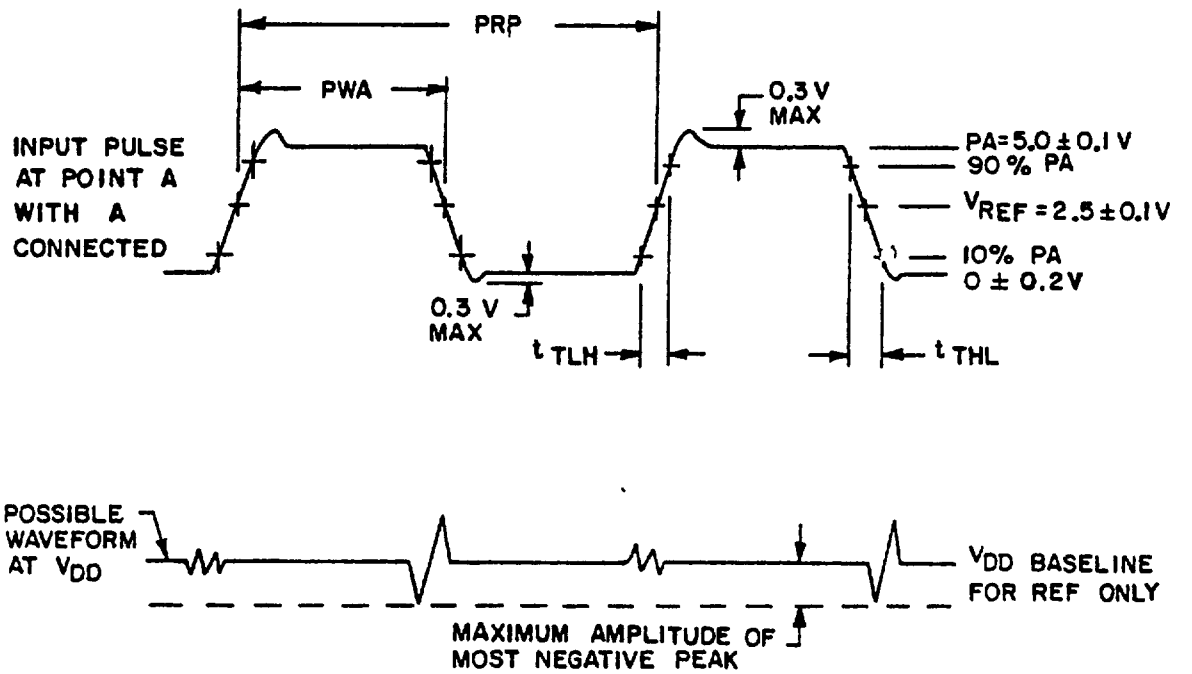


FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

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NOTE:

Input pulse parameters. With the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making the measurement.

$t_{TLH} = t_{THL} = 20 \pm 4$ ns.
PWA = 25 ±2.5 µs.
PRP = 50 ±5.0 µs.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XIX. Power supply filter capacitance test.

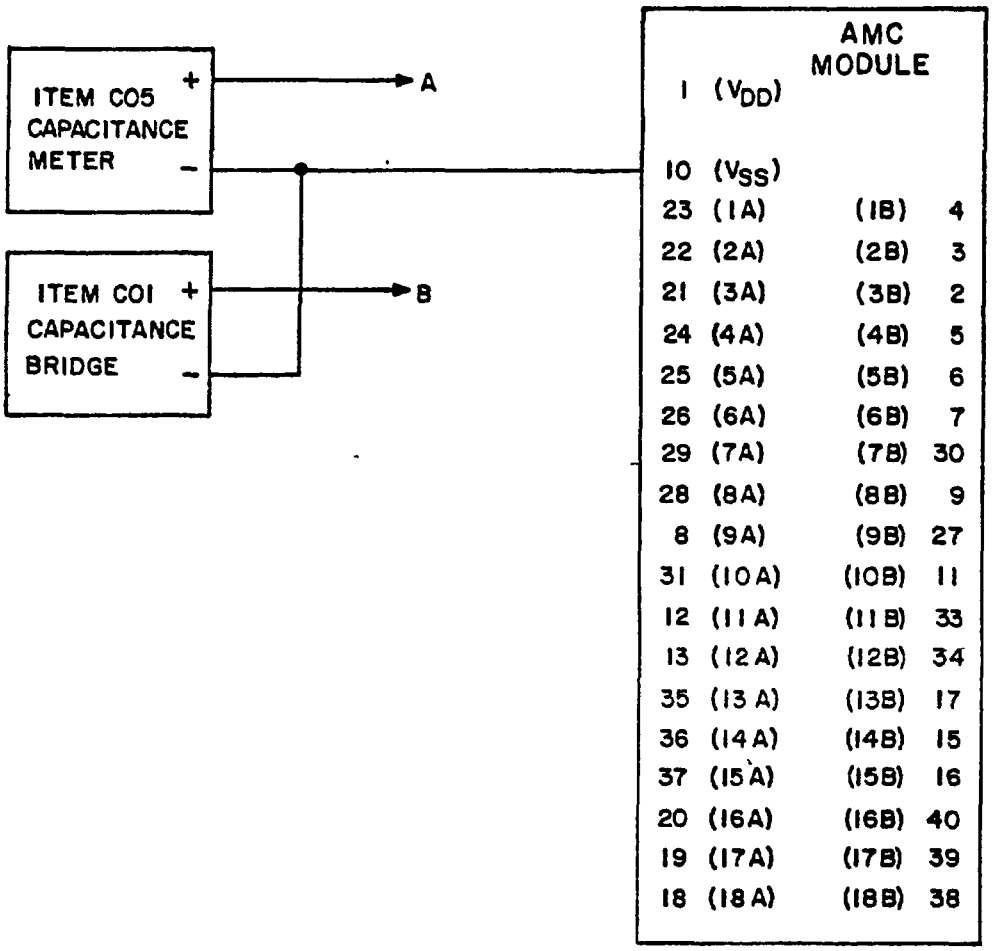
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	0.007	0.013	µF	20

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25 °±5 °C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1A Input capacitance	23	Open	2	25	pF	20
V2	2A	22					
V3	3A	21					
V4	4A	24					
V5	5A	25					
V6	6A	26					
V7	7A	29					
V8	8A	28					
V9	9A	8					
V10	10A	31					
V11	11A	12					
V12	12A	13					
V13	13A	35					
V14	14A	36					
V15	15A	37					
V16	16A	20					
V17	17A	19					
V18	18A	18					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup: Frequency = 1.0 kHz. V_{pk} = 0.3 V maximum.

FIGURE 120. Power supply filter and input capacitance test circuit for tables XIX and XX.

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